

INSTITUTION Electronic Banking Services Policy

The board of directors and management of INSTITUTION recognize that the impact of electronic commerce on the banking industry has grown at a rapid pace. The bank offers innovative electronic products and services designed to meet the needs of its client base. There are significant and unique risks involved in electronic banking services that the bank manages daily.

Therefore, the board adopts the following electronic banking policy. The critical elements of this policy include assessing risk, assigning accountability, active board management, ongoing testing and monitoring, continuing training and education, development of internal controls, and planning for future developments.

ELECTRONIC BANKING SERVICES

Generally, electronic banking services can be segregated into three categories by degree of functionality. First, systems simply provide information (information-only systems), second, they allow users to share information (including personally identifiable information) and communicate with the INSTITUTION (electronic information transfer systems), and third, they facilitate electronic fund transfers and other financial transactions (electronic payment systems).

Information-Only Systems

Information-only systems are those that allow access only to general purpose, marketing, or other publicly available information. The INSTITUTION determines what information is made available, and reviews and updates the content on a regular basis.

Presently, the INSTITUTION's only information-only system is its website: MY INSTITUTION.com.

Electronic Information Transfer Systems

Electronic information transfer systems are those that provide the ability to transmit messages, documents, or files among a group of users. It includes systems in which data or files may be uploaded and downloaded between users and proprietary databases or networks. Privacy, data integrity, and user legitimacy are essential to reducing risk in the operation of information transfer systems.

The INSTITUTION presently offers the following electronic information transfer systems:

- Online banking services, Retail Online Banking and Business Online Banking, accessible via the INSTITUTION's website
- Sharefile
- Fiserv Move-It
- Automatic teller machines (ATM)
- Email
- eStatements

- Mobiliti Mobile Banking (IOS and Android applications)
- Multibank Reporting BAI2

Electronic Payment Systems

Electronic payment systems facilitate electronic fund transfers and other financial transactions the general elements of the payment system are still present (i.e., payment entry, settlement, and distribution). Electronic payment systems can be broadly categorized according to system components, process methodology, and system structure. Risk will vary significantly depending upon the system implementation, administration, and controls used.

The present structure of the INSTITUTION's electronic payment systems is as follows:

- Automated Clearing House (ACH) processing via Business Online Banking ACH Manager module
- Wire transfer payment entry and approval via Business Online Banking Wire Manager module
- Wire transfer settlement and distribution (Payments Exchange: Fedwire, previously known as WireXchange)
- Debit cards/ATM cards (Fiserv EFT)
- Online banking (Retail Online Banking & Business Online Banking)
- CheckFree Bill payment (via Retail Online Banking, Business Online Banking and Mobiliti apps)
- Business Mobiliti, Retail Mobiliti, and Retail Mobiliti Tablet Mobile Banking (IOS and Android applications) and Mobile deposit capture
- Jack Henry Remote Deposit Capture
- Zelle Payments via Retail Online Banking and Retail Mobiliti Application
- TMR Lock Box services

Of the three system categories the level of risk to the INSTITUTION is most significant in the area of electronic payment systems. We have identified ACH processing, wire transfer, bill payment, mobile deposit capture, and remote deposit capture as the largest risk areas for our INSTITUTION. We have built in controls to manage the risk of processing all of these payment systems that are commensurate with the risk involved. Complete details of these risks and the associated controls can be referenced in the INSTITUTION's Electronic Banking Risk Assessment. To process electronic banking payments, the INSTITUTION is dependent on third party vendors that are closely scrutinized via the INSTITUTION's vendor management program.

Electronic Fraud Protection Systems

The INSTITUTION continually monitors fraud trends to ensure we are offering our clients solutions to assist in protecting their accounts.

- Check Positive Pay- Clients have the ability to upload their check issue files and make their positive pay exception decisions within their Business Online Banking or Business Mobiliti app.

- Premium Positive Pay- Clients have the ability to upload their check issue files, monitor ACH activity, and make their positive pay exception decisions within their Business Online Banking.
- ACBT Card Control- Clients have the ability to register their debit cards via the mobility app. Once the card is registered the client could turn the card on/off, set specific limits, or set specific transaction types. The client settings in Card Control do not override INSTITUTION defined card limits and it does not bypass the EnFact card monitoring process.

ASSESSMENT OF RISK TO THE INSTITUTION

Outsourced electronic banking services can contribute to operational risks (also referred to as transaction risks). Operational risk may arise from fraud, error, or the inability to deliver products or services, maintain a competitive position, or manage information. It exists in each process involved in the delivery of the INSTITUTION's products or services. Operational risk not only includes operations and transaction processing, but also areas such as client service, systems development and support, internal control processes, and capacity and contingency planning. Operational risk also may affect other risks such as interest rate, compliance, liquidity, price, strategic, or reputation risk.

Although new technology has provided important benefits to the INSTITUTION and its clients, it also exposes the INSTITUTION to new and different risks. As the INSTITUTION increases its dependency on technology to deliver services and process information, the risk of adverse consequences from operational failure increases. These risks ultimately threaten the safety and soundness of the institution.

The INSTITUTION has identified and documented a comprehensive electronic banking risk assessment and information technology cyber security policy. Since technology growth is not static, new risks are regularly created. The board directs executive management to identify risks to the INSTITUTION by using a process of monitoring, measuring, and identifying risks for which the INSTITUTION must be prepared.

The board appoints the assistant vice president of deposit operations as the individual ultimately responsible for assessing risks and implementing procedures to minimize those risks to the INSTITUTION. It also directs audit personnel to conduct reviews of all areas of electronic banking to assess the internal control structure put in place by INSTITUTION management and to verify that all INSTITUTION departments adhere to the general requirements of this policy.

MITIGATION OF RISK AND RISK MANAGEMENT

The INSTITUTION includes the following elements in its electronic banking risk management plan. For specific details on management's assessment and mitigation of IT and electronic banking risks, reference the Electronic Banking Authentication and Risk Assessment and Information Technology Cyber Security Policy.

Planning and Feasibility Analysis

Ensure that each proposal for new electronic banking services will be analyzed in three stages:

- Study: Needs and objectives will be analyzed and alternatives developed.
- Design and development: The best solution is identified and installed; policies and procedures are developed; and documentation is completed.
- Operation: The system is properly operated and maintained.

The Product Risk Analysis form will be prepared and presented to our Enterprise Risk Management Committee to thoroughly review all new electronic banking services.

Incident Planning and Preparedness

The INSTITUTION maintains a business continuity plan, procedures, and an incident response plan to prepare the INSTITUTION for risks that may occur and outline action plans to respond to disasters and incidents. The business continuity plan includes regular testing of key electronic banking services.

Internal Routines and Controls

Bank management has developed a range of comprehensive system reviews and tests to validate the controls incorporated into each INSTITUTION area for the purpose of protecting hardware and software, proprietary data, and electronic transmissions. Furthermore, management is instructed to train and maintain users on the need to adhere to control standards.

Management Supervision and Internal Controls

Management and the board of directors are actively involved in the decision-making processes to effectively mitigate risks. The INSTITUTION has established controls and processes to mitigate electronic banking risks. In addition, the following oversight is in place:

1. Electronic Banking Risk Assessment
2. IT Committee – Quarterly meetings
3. IT Risk Assessment
4. IT Strategic Plan prepared annually
5. IT Audit – Annual review and testing of controls & supervision
6. IT dashboard
7. Deposit Operations Dashboard
8. Electronic Banking Summary

Reports are provided to the board periodically to detail the actions taken to assess and manage electronic banking risks.

Authentication Controls

The INSTITUTION will require user authentication controls on all electronic banking services used by clients. Different degrees of authentication requirements will be applied in accordance with the perceived risk. Attached to this policy as Exhibit A are the current authentication controls utilized.

Client Enrollment Authentication Controls are covered in the Electronic Banking Authentication and Risk Assessment.

Operations and procedures requiring Dual Control

The INSTITUTION will develop internal processing procedures designed to mitigate the risks of providing electronic banking services. Electronic processes which are higher risk in nature will require dual controls or enhanced system access by employees. These processes have been identified and attached to this policy as Exhibit B.

Vendor Due Diligence

Due to the specialized expertise needed to design, implement, and service new technologies, vendors are needed to provide resources that the INSTITUTION is unable to provide on its own. However, the board and senior management remain responsible for the performance and actions of these vendors while they are performing work for the INSTITUTION. Management will perform due diligence on these vendors as outlined in detail in the Information Technology Cyber Security Policy.

The electronic banking channels are hosted at Fiserv Johns Creek. Part of the annual vendor due diligence completed on Fiserv is a review of their SOC1 and their requirements for encryption of data communications.

Expertise and Training

Employee Education:

It is critical to the success of a risk management program that key employees and vendors have the expertise and skills necessary to perform critical functions, and that they are and remain properly trained. Management will allocate sufficient resources to hire and maintain employee training and to ensure that adequate backup exists if critical people leave. Training will include technical course work, attendance at industry conferences, participation in industry working groups, as well as time allotments for appropriate staff to keep abreast of important technological and market developments.

Client Education:

Training and resource guides are given to clients upon onboarding. Annually, additional education and Self Certification, in areas where applicable is completed. Periodically the INSTITUTION will send out a communication piece regarding phishing and other online fraud schemes.

Testing

Testing of systems is necessary to validate that equipment and systems function properly and produce the desired results. Management will verify whether new systems operate effectively with the INSTITUTION's existing technology and will include vendors in the testing process.

Exhibit A Authentication Controls on Electronic Banking Services Access for Clients

Application Access	Client Control
Business Online Banking (including CheckFree Business Bill Pay, Business Mobiliti and Business Mobile Deposit Capture, Positive Pay, and Positive Pay Premium) – General Controls	• User ID and Password/Token, required for System Access;. (Tokens required for some access levels) • Client user administration may be controlled by ACBT if Self-Administration agreement is not on file. (i.e., only INSTITUTION employees have the ability to create a new User or change User rights) • Client user administration with a Self-Administration agreement on file (i.e., only assigned clients may create or change user rights) • Recommended security features: • IP restrictions • Day/time restrictions Business Mobiliti is a user level setting. Client requests the access be enabled.
Business Online Banking – ACH Origination	• User ID, Password/ Token required for System Access • Clients that originate > \$50k it is recommended they have dual control. • Client exposure limits assigned by INSTITUTION for clients that are not operating under a Pre-Fund model.

	• Users may be assigned limits < the Client limit • File creation may be limited to pre-defined templates
Business Online Banking - Wire Transfer	• User ID, Password/Token required for System Access • Dual control or “offline” approval via callback with PIN required • Daily or periodic limits may be assigned for each Client • Users may be assigned limits < the Client limit • File creation is limited to pre-defined templates
Retail Online Banking (including Mobiliti mobile banking, Mobiliti mobile deposit capture, Positive Pay, Mobiliti Tablet, and CheckFree Bill Pay/Zelle)	• User ID and Password • Challenge questions • Mobile device registration • CheckFree is not activated until new enrollment is verified
Remote Deposit Capture	• User ID & Password for System Access with Jack Henry (Profit Stars) • Client user administration is controlled by ACBT (i.e., only INSTITUTION employees have the ability to create a new User or change User rights) • Client user administration with a Self-Administration agreement on file (i.e., only assigned clients may create or change user rights)

Exhibit B
Internal Electronic Processes Which Require Enhanced System Access or Dual Processing Control

Process	Control
Payments Exchange: Fedwire wire processing	• User ID, Passcode, & Token required for System Access • Additional PIN required to release outgoing wires • Access limited to registered IP addresses • Day/time restrictions • Dual control – two employees required to originate a wire
WIRE PROCESSING TSP processing, foreign dollar and wires not able to be processed via Payments Exchange due to lack of correspondent INSTITUTION information.	• User ID, Passcode, & Token required for System Access • Additional PIN required to release outgoing wires • Dual control – two employees required to originate a wire
SWIFT Alliance Lite2	• SWIFT code used to receive foreign wires • Alliance Lite2 software used to send messages to other SWIFT participants • Actual movement of funds happens through Payments Exchange: Fedwire platform
Item Processing – Inbound & Outbound Cash Letter Processing, adjustments, and ACH Returns	• User ID, Password & Token Required for System Access • Federal Reserve Bank security certificate required
Fiserv EFT (Debit Card Maintenance and ENFACT)	• User ID, PIN and Token Required for System Access